

STAT 440/540 Milestone 2: EDA & Initial Models

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Table of contents

1 Overview	1
1.1 Package Setup	1
1.2 Data Import and Cleaning	1
1.3 Descriptive Statistics	2
1.4 Exploratory Visualizations	4
1.5 Initial Simple Regressions	5
1.6 Early Findings and Next Steps	6
1.7 Reproducibility Snapshot	6

1 Overview

- Briefly restate question, dataset, and key variables.

The purpose of the this project is to explore the relationship that mental health and other life style factors that can affect a college student's GPA. This dataset gathers data on GPA, mental health factors, student habits, and other life style factors. The key variables in this data set is the repsonse variable *GPA* and other important variables such as sleep and [something else].

1.1 Package Setup

```
if (!requireNamespace("pacman", quietly = TRUE)) install.packages("pacman")
pacman::p_load(tidyverse, janitor, skimr, GGally, broom, patchwork, knitr, kableExtra)
set.seed(440540)
```

1.2 Data Import and Cleaning

```
raw <- readr::read_csv("../data/raw/student_lifestyle_dataset.csv") |> janitor::clean_names()

# raw <- readr::read_csv("../data/raw/example_stub.csv") |> janitor::clean_names()

summary(raw)
```

```

  student_id      study_hours_per_day extracurricular_hours_per_day
Min.   :   1.0    Min.   : 5.000      Min.   :0.00
1st Qu.: 500.8    1st Qu.: 6.300      1st Qu.:1.00
Median :1000.5    Median : 7.400      Median :2.00
Mean   :1000.5    Mean   : 7.476      Mean   :1.99
3rd Qu.:1500.2    3rd Qu.: 8.700      3rd Qu.:3.00
Max.   :2000.0    Max.   :10.000     Max.   :4.00
sleep_hours_per_day social_hours_per_day physical_activity_hours_per_day
Min.   : 5.000      Min.   :0.000      Min.   : 0.000
1st Qu.: 6.200      1st Qu.:1.200      1st Qu.: 2.400
Median : 7.500      Median :2.600      Median : 4.100
Mean   : 7.501      Mean   :2.705      Mean   : 4.328
3rd Qu.: 8.800      3rd Qu.:4.100      3rd Qu.: 6.100
Max.   :10.000      Max.   :6.000      Max.   :13.000
  gpa      stress_level
Min.   :2.240    Length:2000
1st Qu.:2.900    Class :character
Median :3.110    Mode  :character
Mean   :3.116
3rd Qu.:3.330
Max.   :4.000
```

```
# dat <- raw

dat <- raw |>
  dplyr::select(-student_id)

# Code originally given
# mutate(
#   x3 = factor(x3)
# ) |>
# tidyr::drop_na(y)
```

1.3 Descriptive Statistics

```

summary_data <- dat |>
  dplyr::select(where(is.numeric)) |>
  skimr::skim()

# First half of kable
summary_data |>
  dplyr::select(
    skim_type,
    skim_variable,
    n_missing,
    complete_rate,
    numeric.mean,
    numeric.sd,
  ) |>
  knitr::kable()
# Second half of kable
summary_data |>
  dplyr::select(
    skim_variable,
    numeric.p0,
    numeric.p25,
    numeric.p50,
    numeric.p75,
    numeric.p100
  ) |>
  knitr::kable()
# THE original CODE GIVEN
# dat |>
#   dplyr::select(where(is.numeric)) |>
#   skimr::skim() |>
#   knitr::kable()

```

Table 1: Summary statistics for numeric variables.

skim_type	skim_variable	n_missing	complete_rate	numeric.mean	numeric.sd
numeric	study_hours_per_day	0	1	7.47580	1.4238884
numeric	extracurricular_hours_per_day	0	1	1.99010	1.1558547
numeric	sleep_hours_per_day	0	1	7.50125	1.4609485
numeric	social_hours_per_day	0	1	2.70455	1.6885141
numeric	physical_activity_hours_per_day	0	1	4.32830	2.5141101
numeric	gpa	0	1	3.11596	0.2986735

Table 2: Summary statistics for numeric variables.

skim_variable	numeric.p0	numeric.p25	numeric.p50	numeric.p75	numeric.p100
study_hours_per_day	5.00	6.3	7.40	8.70	10
extracurricular_hours_per_day	0.00	1.0	2.00	3.00	4
sleep_hours_per_day	5.00	6.2	7.50	8.80	10
social_hours_per_day	0.00	1.2	2.60	4.10	6
physical_activity_hours_per_day	0.00	2.4	4.10	6.10	13
gpa	2.24	2.9	3.11	3.33	4

1.4 Exploratory Visualizations

```
# ---- From old data set
# gpa_dat <- dat |>
#   dplyr::select(academic_performance_gpa, study_hours_per_week, financial_stress)

# gpa_dat2 <- dat |>
#   dplyr::select(academic_performance_gpa, mental_stress_level, diet_quality, cognitive_distor

# ----

GGally::ggpairs(dplyr::select(dat, where(is.numeric)))
```

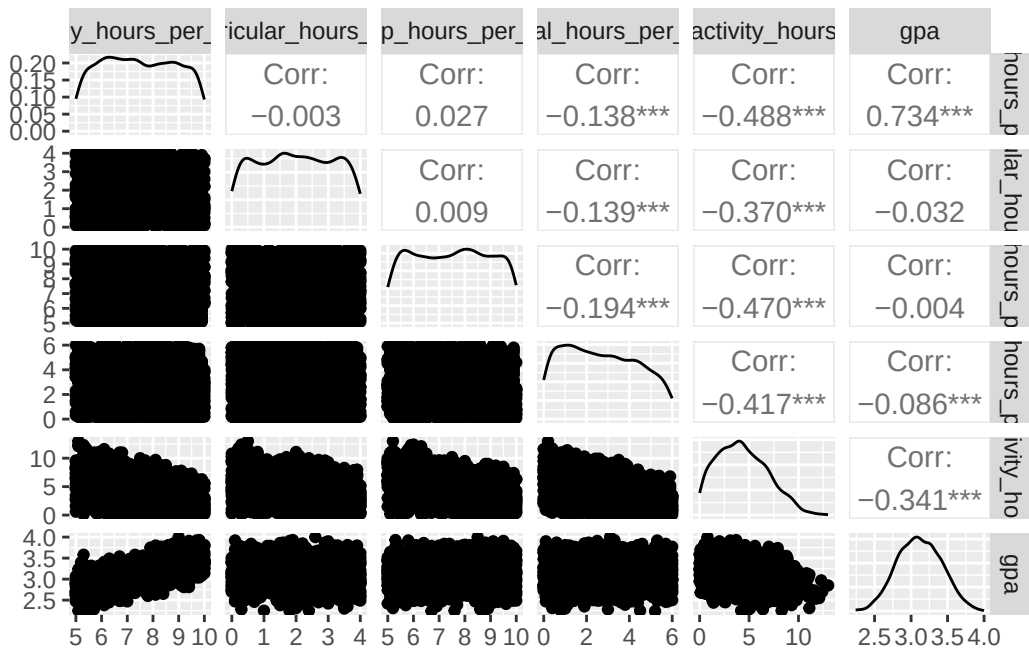


Figure 1: Pairs plot for numeric variables.

```

dat |>
  dplyr::select(where(is.numeric)) |>
  tidyr::pivot_longer(cols = -c(gpa), names_to = "var", values_to = "val") |>
  ggplot(aes(val, gpa)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "loess", se = TRUE) +
  facet_wrap(~ var, scales = "free_x")

# dat |>
# dplyr::select(where(is.numeric)) |>
# tidyr::pivot_longer(cols = -c(y), names_to = "var", values_to = "val") |>
# ggplot(aes(val, y)) +
# geom_point(alpha = 0.5) +
# geom_smooth(method = "loess", se = TRUE) +
# facet_wrap(~ var, scales = "free_x")

```

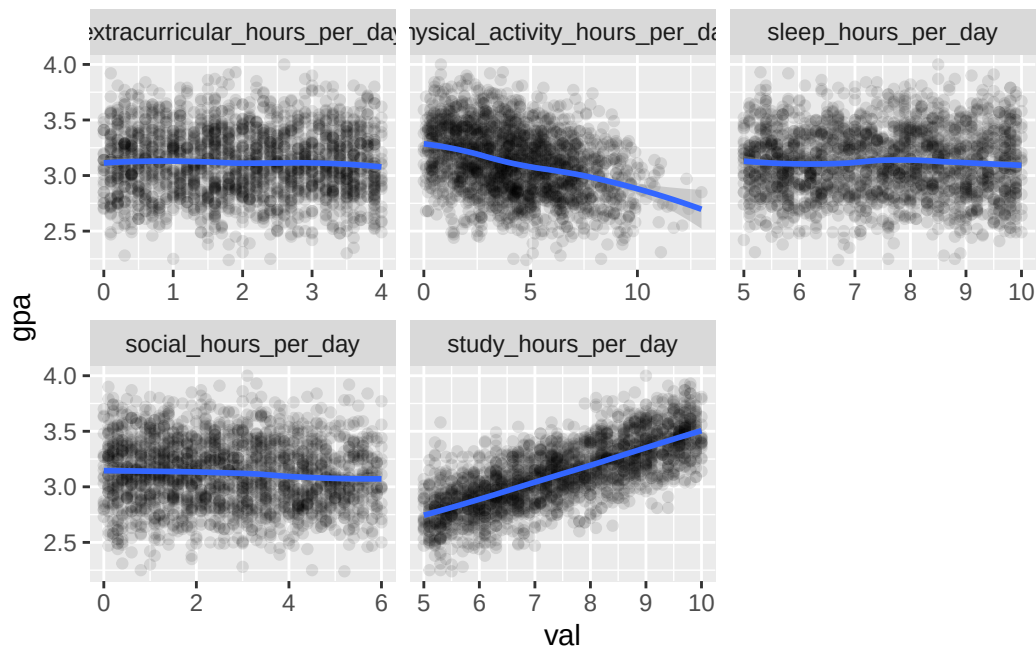


Figure 2: Outcome vs. selected predictors with loess smooth.

1.5 Initial Simple Regressions

```

cand_x <- c("study_hours_per_day", "extracurricular_hours_per_day", "sleep_hours_per_day", "social_hours_per_day")
fits <- lapply(cand_x, \(v) lm(reformulate(v, response = "gpa"), data = dat))
models <- tibble::tibble(
  predictor = cand_x,
  glance = purrr::map(fits, broom::glance),

```

```

  tidy = purrr::map(fits, broom::tidy)
)

dplyr::bind_rows(purrr::map2(models$predictor, models$glance, \(v,g) dplyr::mutate(g, predictor = v)) |>
  dplyr::select(predictor, r.squared, adj.r.squared, AIC, BIC, p.value) |>
  knitr::kable(digits = 3, caption = "Simple regression comparisons")

```

Table 3: Simple regression comparisons

predictor	r.squared	adj.r.squared	AIC	BIC	p.value
study_hours_per_day	0.539	0.539	-703.501	-686.698	0.000
extracurricular_hours_per_day	0.001	0.001	845.066	861.869	0.150
sleep_hours_per_day	0.000	0.000	847.101	863.903	0.848
social_hours_per_day	0.007	0.007	832.402	849.205	0.000
physical_activity_hours_per_day	0.116	0.116	599.670	616.472	0.000

```

# cand_x <- c("x1","x2","x3")
# fits <- lapply(cand_x, \(v) lm(reformulate(v, response = "y"), data = dat))
# models <- tibble::tibble(
#   predictor = cand_x,
#   glance = purrr::map(fits, broom::glance),
#   tidy = purrr::map(fits, broom::tidy)
# )

# dplyr::bind_rows(purrr::map2(models$predictor, models$glance, \(v,g) dplyr::mutate(g, predictor = v)) |>
#   dplyr::select(predictor, r.squared, adj.r.squared, AIC, BIC, p.value) |>
#   knitr::kable(digits = 3, caption = "Simple regression comparisons")

```

1.6 Early Findings and Next Steps

- What looks promising? Any evidence of nonlinearity or interactions to explore?

1.7 Reproducibility Snapshot

```
sessionInfo()
```

R version 4.5.0 (2025-04-11)

Platform: aarch64-apple-darwin20

Running under: macOS Sonoma 14.3

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LA

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Denver

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] kableExtra_1.4.0 knitr_1.50 patchwork_1.3.0 broom_1.0.8
[5] GGally_2.2.1 skimr_2.2.1 janitor_2.2.1 lubridate_1.9.4
[9] forcats_1.0.0 stringr_1.5.1 dplyr_1.1.4 purrr_1.0.4
[13] readr_2.1.5 tidyr_1.3.1 tibble_3.2.1 ggplot2_3.5.2
[17] tidyverse_2.0.0

loaded via a namespace (and not attached):

[1] gtable_0.3.6 xfun_0.52 lattice_0.22-6 tzdb_0.5.0
[5] vctr_0.6.5 tools_4.5.0 generics_0.1.4 parallel_4.5.0
[9] pacman_0.5.1 pkgconfig_2.0.3 Matrix_1.7-3 RColorBrewer_1.1-3
[13] lifecycle_1.0.4 compiler_4.5.0 farver_2.1.2 textshaping_1.0.1
[17] repr_1.1.7 snakecase_0.11.1 htmltools_0.5.8.1 yaml_2.3.10
[21] pillar_1.10.2 crayon_1.5.3 nlme_3.1-168 ggstats_0.9.0
[25] tidyselect_1.2.1 digest_0.6.37 stringi_1.8.7 labeling_0.4.3
[29] splines_4.5.0 fastmap_1.2.0 grid_4.5.0 cli_3.6.5
[33] magrittr_2.0.3 base64enc_0.1-3 withr_3.0.2 scales_1.4.0
[37] backports_1.5.0 bit64_4.6.0-1 timechange_0.3.0 rmarkdown_2.29
[41] bit_4.6.0 hms_1.1.3 evaluate_1.0.3 viridisLite_0.4.2
[45] mgcv_1.9-1 rlang_1.1.6 Rcpp_1.0.14 glue_1.8.0
[49] xml2_1.3.8 svglite_2.2.1 rstudioapi_0.17.1 vroom_1.6.5
[53] jsonlite_2.0.0 R6_2.6.1 plyr_1.8.9 systemfonts_1.2.3